

Supplementary Material for Spectral Radius Minimization for Signed Squares of Cycles

Anonymous

Abstract

This supplement records two exact components of the proof package. First, it proves that a finite signed ring with $r \in \{1, 2, 3\}$ mutually separated elementary six-gap interfaces has exactly $2r$ squared eigenvalues in a fixed neighborhood of the one-interface level c_6 , provided the minimum cyclic separation is at least 1040. The proof includes the finite-ring patch identification, the full codimension- $2r$ complement estimate, and the $2r$ -dimensional Feshbach equation. Second, it gives all integer Rayleigh witnesses used to separate every other abnormal positive single gap from the six-gap edge by more than $1/250$. The final section identifies the exact repository artifacts and independent checkers supporting the computer-assisted inputs. All finite-support comparisons printed here are checkable by integer arithmetic.

Contents

1	Exact $2r$ clusters for separated interfaces	2
1.1	Operators and the one-interface input	2
1.2	Finite-ring geometry and patch identification	2
1.3	Tails, Gram matrix, and the lower count	4
1.4	The codimension- $2r$ complement	5
1.5	The $2r$ -dimensional Feshbach equation	5
1.6	A sufficient exponential onset	6
2	Complete integer witnesses for abnormal single gaps	7
2.1	Canonical model and comparison threshold	7
2.2	Six small-gap witnesses	8
2.3	One witness for every $g \geq 9$	9
2.4	Completion of the single-gap hierarchy	9
3	Reproducibility manifest	10
3.1	Checkpoint and evidence boundary	10
3.2	Exact artifact inventory	10
3.3	Verification commands	12
3.4	Scope of the supplement	13

1 Exact $2r$ clusters for separated interfaces

1.1 Operators and the one-interface input

Let $Q^{(6)} \in \{\pm 1\}^{\mathbb{Z}}$ be the flux word whose positive sites are

$$\{-4j : j \geq 0\} \cup \{6 + 4j : j \geq 0\}.$$

Choose a lift $\tau^{(6)}$ with $\tau_{i+1}^{(6)} = Q_i^{(6)} \tau_i^{(6)}$, and define

$$(A_6 u)_i = u_{i-1} + u_{i+1} + \tau_{i-2}^{(6)} u_{i-2} + \tau_i^{(6)} u_{i+2}, \quad H_6 = A_6^2. \quad (1)$$

The other lift is $-\tau^{(6)}$; after squaring it is diagonally unitarily equivalent to the displayed operator. Reflection gives the reverse orientation.

Define

$$\begin{aligned} p_6(y) = & 16y^{10} - 520y^9 + 6913y^8 - 48448y^7 + 191768y^6 - 423904y^5 \\ & + 484528y^4 - 270464y^3 + 137856y^2 - 19968y + 256. \end{aligned} \quad (2)$$

The constant c_6 is the unique root of p_6 in the rational interval

$$\frac{7905369311620327}{10^{15}} < c_6 < \frac{7905369311620328}{10^{15}}. \quad (3)$$

The period-eight reference squared edge is

$$\eta = 4 + \sqrt{10 + 2\sqrt{5}} < c_6. \quad (4)$$

The proof below uses the following certified one-interface and Floquet inputs. They are the only computer-assisted spectral inputs to this section:

$$\ker(H_6 - c_6) = \text{span}\{\psi_+, \psi_-\}, \quad \dim \ker(H_6 - c_6) = 2, \quad (5)$$

$$H_6|_{\ker(H_6 - c_6)^\perp} \leq c_6 - \frac{1}{100}, \quad (6)$$

where the modes are orthonormal and satisfy $A_6 \psi_\pm = \pm \sqrt{c_6} \psi_\pm$. For all eight cuts of the period-eight bulk, both stable multipliers have modulus below

$$q_F = \frac{9}{25}, \quad (7)$$

and the selected two-column Floquet bases have condition number below 17. These bounds hold for both tails and are uniform under reflection.

1.2 Finite-ring geometry and patch identification

Work on $\mathbb{Z}/n\mathbb{Z}$ in Hamilton gauge. All step-one signs are positive except for a movable seam carrying a holonomy $\alpha \in \{+1, -1\}$. The flux word is $Q_i = \tau_i \tau_{i+1}$. Assume that its positive sites have cyclic gaps four except at exactly $r \in \{1, 2, 3\}$ marked gaps of length six. Write p_j for the left positive endpoint of the j -th marked gap, ordered cyclically, and put

$$d_j = p_{j+1} - p_j \pmod{n}, \quad D = \min_j d_j, \quad (8)$$

with $p_{r+1} = p_1$. For $r = 1$, the return arc has length $d_1 = D = n$. Thus D is a distance in sites, not a number of period-eight cells. Assume from now on that

$$D \geq 1040, \quad S = \left\lfloor \frac{D}{4} \right\rfloor, \quad a = S - 8. \quad (9)$$

On the oriented arc from p_j to p_{j+1} , use the coordinate $0 \leq t \leq d_j$. Define interface cutoffs χ_j and a bulk cutoff β_j by

range of t	χ_j	β_j	$\vartheta = \frac{\pi(t-a)}{2S},$
$0 \leq t \leq a$	1	0	
$a < t < a + S$	$\cos \vartheta$	$\sin \vartheta$	
$a + S \leq t \leq d_j - a - S$	0	1	

(10)

and on the right transition by

$$\beta_j = \cos \varphi, \quad \chi_{j+1} = \sin \varphi, \quad \varphi = \frac{\pi(t - d_j + a + S)}{2S} \quad (11)$$

for $d_j - a - S < t < d_j - a$, followed by $\chi_{j+1} = 1$ on $d_j - a \leq t \leq d_j$. Endpoint values agree. For $r = 1$, the two endpoint pieces are the same cyclic interface cutoff. Pointwise,

$$\sum_{j=1}^r \chi_j^2 + \sum_{j=1}^r \beta_j^2 = 1. \quad (12)$$

The pure-bulk plateau on the j -th arc is $[p_j + 2S - 8, p_{j+1} - 2S + 8]$, and its length is at least

$$d_j - 4S + 16 \geq D - 4\lfloor D/4 \rfloor + 16 \geq 16. \quad (13)$$

For a set E , let $E^{[k]}$ denote its cyclic k -neighborhood. Set

$$I_j = \{i : \chi_j(i) \neq 0\}, \quad J_j = I_j^{[4]}, \quad B_j = \{i : \beta_j(i) \neq 0\}, \quad C_j = B_j^{[4]}.$$

Since $H = A^2$ has range four, the coefficient collar needed to identify a compression to J_j is contained in $I_j^{[6]}$. That collar stays at distance at least

$$D - 2\lfloor D/4 \rfloor - 4 \geq 516 \quad (14)$$

from every other marked core. Likewise $B_j^{[6]}$ stays at distance at least $S - 19 \geq 241$ from each six-gap core. Hence every interface patch contains exactly one marked core and every bulk patch is a pure period-eight patch.

The holonomy seam can be moved by diagonal switching. Place it on an edge of a bulk plateau whose endpoints have distance at least eight from every interface support. It then misses every $I_j^{[6]}$. Each interface collar is a proper arc, so after unwrapping it, translation, optional reflection, and diagonal switching identify its coefficients exactly with the corresponding finite section of H_6 . On a proper arc, two lifts of the same Q -word differ by a global sign. If $(D_0 u)_i = (-1)^i u_i$, then

$$A_{-\tau} = -D_0 A_\tau D_0, \quad H_{-\tau} = D_0 H_\tau D_0. \quad (15)$$

This removes the remaining lift choice after squaring. The same argument identifies each C_j with a finite section of the period-eight reference operator. We record the result in the form needed below.

Lemma 1.1 (Patch identification). *For every allowed orientation, lift, and holonomy, there are local unitaries such that:*

- (i) *the compression of H to J_j is a principal finite section of the corresponding infinite $G6$ operator;*
- (ii) *the image under H of a vector supported in I_j is supported in J_j and agrees with the image under that infinite operator;*
- (iii) *the analogous compression and map on $B_j \subset C_j$ agree with a period-eight reference-bulk finite section.*

These statements include cyclic wraparound because every collar is first unwrapped as a proper arc.

Transport the orthonormal pair ψ_+, ψ_- through the same local unitary, before truncation, and call the transported modes $\psi_{j,+}, \psi_{j,-}$. For every finite-ring vector x , zero extension on the unwrapped arc gives

$$\langle \psi_{j,\pm}, \chi_j x \rangle_{\ell^2(\mathbb{Z})} = \langle \chi_j \psi_{j,\pm}, x \rangle_{\mathbb{C}^n}. \quad (16)$$

This identity is the bridge that permits the complete rank-two complement bound (6) to be used on each localized vector.

1.3 Tails, Gram matrix, and the lower count

The cutoff starts changing at distance $S - 8$ from a marked core. Removing the propagation range four leaves

$$L_{\text{site}} = S - 12, \quad \ell = \left\lfloor \frac{L_{\text{site}}}{8} \right\rfloor. \quad (17)$$

The eight site residues, the two tails, the condition-number bound, and the geometric series imply, for either normalized local mode,

$$\|\psi_{j,\pm}\|_{\text{tail}}^2 \leq \frac{16 \cdot 17^2}{1 - q_F^2} q_F^{2\ell} = \frac{10625}{2} q_F^{2\ell} < 73^2 q_F^{2\ell}. \quad (18)$$

Put

$$\phi_{j,\pm} = \chi_j \psi_{j,\pm}, \quad \Phi = (\phi_{1,+}, \phi_{1,-}, \dots, \phi_{r,+}, \phi_{r,-}), \quad G = \Phi^* \Phi. \quad (19)$$

The full modes at one interface are orthogonal. Their truncated cross term is not discarded; rather,

$$|\langle \phi_{j,+}, \phi_{j,-} \rangle| \leq 73^2 q_F^{2\ell}.$$

The same estimate holds between different interfaces because both vectors are then in their tails. With $m = 2r$,

$$\|G - I_m\| \leq m 73^2 q_F^{2\ell}. \quad (20)$$

The hypotheses give $S \geq 260$, $L_{\text{site}} \geq 248$, and $\ell \geq 31$. Since $m \leq 6$,

$$6 \cdot 73^2 \left(\frac{9}{25} \right)^{62} < \frac{1}{2}. \quad (21)$$

Thus the $2r$ columns are independent and $\|G^{-1}\| \leq 2$.

The row-sum bound gives $\|H\| \leq 16$, while $c_6 < 8$. Cutting an exact mode only in its tails and using (18) yields

$$\|(H - c_6)\phi_{j,\pm}\| \leq (16 + 8) 73 q_F^\ell = 1752 q_F^\ell. \quad (22)$$

Define the isometry

$$U = \Phi G^{-1/2} : \mathbb{C}^{2r} \longrightarrow \mathbb{C}^n. \quad (23)$$

Then

$$\|(H - c_6)U\| < 3504rq_F^\ell < \frac{1}{400}. \quad (24)$$

Let E_W be the spectral projection of H onto

$$W = \left[c_6 - \frac{1}{400}, c_6 + \frac{1}{400} \right]. \quad (25)$$

If $\text{rank } E_W < 2r$, a nonzero vector in $\text{ran } U$ is orthogonal to $\text{ran } E_W$. The spectral theorem then makes its residual at least $1/400$, contradicting (24). Therefore

$$\text{rank } E_W \geq 2r. \quad (26)$$

1.4 The codimension- $2r$ complement

Let $V = \text{ran } \Phi$. If $x \perp V$, then (16) gives

$$\langle \psi_{j,+}, \chi_j x \rangle = \langle \psi_{j,-}, \chi_j x \rangle = 0$$

for every interface. Thus each interface-localized vector is orthogonal to the entire two-dimensional space in (5), and (6) applies. Each bulk-localized vector satisfies the stronger upper bound $\eta < c_6 - 1/100$.

The cutoff vector follows a unit-circle quarter arc at angular speed $\pi/(2S)$. If two sites have cyclic distance $h \leq 4$, the chord bound gives

$$\sum_{\gamma} |\gamma(u) - \gamma(v)|^2 \leq \frac{\pi^2 h^2}{4S^2},$$

where γ ranges over all cutoffs. The exact double-commutator IMS identity, the range-four row-sum bound, and $\pi^2 < 10$ consequently give

$$\|E_{\text{IMS}}\| < \frac{320}{S^2} \leq \frac{320}{260^2} = \frac{4}{845}. \quad (27)$$

Using (12) and summing the localized quadratic-form bounds, we obtain on $V^\perp$

$$H|_{V^\perp} \leq c_6 - \frac{1}{100} + \frac{4}{845} = c_6 - \frac{89}{16900} < c_6 - \frac{1}{200}. \quad (28)$$

By the codimension form of min-max, at most $\dim V = 2r$ eigenvalues lie above $c_6 - 1/200$. Since the entire window W lies above this level, (26) sharpens to

$$\text{rank } E_W = 2r. \quad (29)$$

The equality counts algebraic multiplicity and makes no assertion of individual simplicity.

1.5 The $2r$ -dimensional Feshbach equation

Let

$$P = UU^*, \quad Q_\perp = I - P, \quad E = (H - c_6)\Phi. \quad (30)$$

Because $\text{ran } U = V$, estimate (28) holds on $\text{ran } Q_\perp$. If $|z - c_6| \leq 1/400$, then

$$\|(Q_\perp H Q_\perp - z)^{-1}\| \leq 400 \quad \text{on } \text{ran } Q_\perp. \quad (31)$$

Block Gaussian elimination with respect to $P + Q_\perp = I$ gives the effective operator on $\mathbb{C}^{2r}$,

$$H_{\text{eff}}(z) = U^* H U - U^* H Q_\perp (Q_\perp H Q_\perp - z)^{-1} Q_\perp H U, \quad (32)$$

and the finite-ring eigenvalue equation in W is exactly

$$\det (H_{\text{eff}}(z) - z I_{2r}) = 0. \quad (33)$$

Using $U = \Phi G^{-1/2}$ and $Q_\perp H U = Q_\perp E G^{-1/2}$, one obtains

$$\begin{aligned} H_{\text{eff}}(z) - c_6 I_{2r} &= G^{-1/2} \Phi^* E G^{-1/2} \\ &\quad - G^{-1/2} E^* Q_\perp (Q_\perp H Q_\perp - z)^{-1} Q_\perp E G^{-1/2}. \end{aligned} \quad (34)$$

No orthogonality of the uncorrected columns of Φ is assumed. Writing the right side as $T_1 + R_2(z)$, the Gram and residual bounds give

$$\|T_1\| \leq 3504 r q_F^\ell, \quad \|R_2(z)\| \leq 400 r 3504^2 q_F^{2\ell}. \quad (35)$$

Since

$$400 \cdot 3504^2 \left(\frac{9}{25}\right)^{31} < 1, \quad (36)$$

we have $\|R_2(z)\| < r q_F^\ell$. Every root of (33) in W therefore satisfies

$$|\lambda_j - c_6| < 3505 r \left(\frac{9}{25}\right)^\ell, \quad 1 \leq j \leq 2r. \quad (37)$$

Theorem 1.2 (Separated-interface cluster). *Let a legal finite ring contain exactly $r \in \{1, 2, 3\}$ elementary $G6$ interfaces, with arbitrary interface orientations, either lift, and either Hamilton holonomy. If their minimum cyclic site separation satisfies $D \geq 1040$, define S, L_{site}, ℓ by (9) and (17). Then*

$$\text{rank } \mathbf{1}_{[c_6-1/400, c_6+1/400]}(H) = 2r.$$

The $2r$ eigenvalues in this window, counted with multiplicity, satisfy (37), and their exact finite-ring equation is the $2r$ -dimensional determinant (33).

1.6 A sufficient exponential onset

The explicit constants also give a continuous sufficient onset for the exponential construction. For every even n ,

$$\rho_-(n)^2 > 8 - \frac{200}{n^2}.$$

For the nonzero residues put

$$B(n) = 8 - \frac{200}{n^2} - \frac{9}{100}.$$

If the corresponding residue construction has r interfaces and minimum separation $D \geq 1040$, the cluster bound gives

$$\rho(A_n)^2 \leq c_6^{\text{up}} + 3505 r \left(\frac{9}{25}\right)^\ell, \quad \ell = \left\lfloor \frac{\lfloor D/4 \rfloor - 12}{8} \right\rfloor,$$

where $c_6^{\text{up}} = 7905369311620328/10^{15}$. The first distance-eligible endpoints are

$n \bmod 8$	n	r	D	ℓ
2	1042	1	1042	31
4	2084	2	1042	31
6	3126	3	1042	31

and exact rational evaluation gives

$$B(n) - \left[c_6^{\text{up}} + 3505r \left(\frac{9}{25} \right)^{31} \right] > \frac{1}{250}$$

in all three rows. Along each residue subsequence, D and ℓ do not decrease, the spectral cap does not increase, and $B(n)$ strictly increases.

For residue zero, the period-eight cap is $1561/200$. At $n = 3120$,

$$8 - \frac{200}{3120^2} = \frac{389375}{48672}, \quad \frac{389375}{48672} - \frac{1561}{200} = \frac{237251}{1216800} > 0.$$

The first four even orders at or above 3120 are 3120, 3122, 3124, 3126, one in each residue class. The preceding endpoint checks and residue-wise monotonicity therefore prove a strict counterexample for every even $n \geq 3120$. We record

$$N_{\text{exp}} = 3120$$

as a sufficient, nonoptimal onset for this exponential estimate. It is not the sharp onset: the finite bridge plus IMS theorem in the main paper gives the stronger value 48.

The restrictions $r \leq 3$ and $D \geq 1040$ enter, respectively, in the uniform Gram estimate and in the seam-clearance, tail, and IMS bounds. The argument proves neither individual simplicity nor a nonzero leading interaction coefficient.

2 Complete integer witnesses for abnormal single gaps

2.1 Canonical model and comparison threshold

For an integer $g \geq 1$, set

$$D_g = \{-4j : j \geq 0\} \cup \{0, g\} \cup \{g + 4j : j \geq 0\}.$$

Let $Q_i = +1$ for $i \in D_g$ and $Q_i = -1$ otherwise. Choose either lift $\tau_{i+1} = Q_i \tau_i$, and define on $\ell^2(\mathbb{Z})$

$$(A_g v)_k = v_{k-1} + v_{k+1} + \tau_{k-2} v_{k-2} + \tau_k v_{k+2}, \quad H_g = A_g^2. \quad (38)$$

Gap four is the unperturbed period-eight bulk. Gap six is the elementary interface of section 1.

For a finitely supported integer vector v , extend it by zero. Since A_g is self-adjoint,

$$\frac{\langle v, H_g v \rangle}{\langle v, v \rangle} = \frac{\|A_g v\|^2}{\|v\|^2}. \quad (39)$$

The certified upper endpoint in (3) shows that it is enough to exceed the rational threshold

$$T = \frac{7905369311620328}{10^{15}} + \frac{1}{250} = \frac{988671163952541}{125000000000000}. \quad (40)$$

For integers $N, D > 0$, define the exact cross-multiplication

$$\Delta(N, D) = 125000000000000N - 988671163952541D. \quad (41)$$

Then $N/D > T$ if and only if $\Delta(N, D) > 0$.

2.2 Six small-gap witnesses

Coordinates in each vector are listed in increasing order on the interval I . The displayed image is listed on $J = I + [-2, 2]$. Every entry follows directly from the four-term formula (38).

Gap $g = 1$. On $I = [-2, 3]$, take

$$\begin{aligned} v &= (2, 0, 4, 4, 6, 5), \\ A_1 v &= (-2, 2, 4, 2, 12, 15, 13, 10, 11, -5). \end{aligned}$$

Then $D = \|v\|^2 = 97$ and $N = \|A_1 v\|^2 = 812$.

Gap $g = 2$. On $I = [-4, 6]$, take

$$\begin{aligned} v &= (1, -1, -2, -1, -4, -5, -6, -2, 1, 4, 2), \\ A_2 v &= (-1, 0, 1, 0, -7, 0, -14, -11, -12, -14, 10, 5, 5, -2, 2). \end{aligned}$$

Then $D = 109$ and $N = 866$.

Gap $g = 3$. On $I = [-5, 8]$, take

$$\begin{aligned} v &= (0, 1, -1, -3, 0, -6, -8, -6, -10, -6, -8, -6, 1, -3), \\ A_3 v &= (0, -1, 0, 2, -2, -8, 0, -17, -22, -18, -28, -18, -23, -16, \\ &\quad -1, -5, -4, 3). \end{aligned}$$

Then $D = 393$ and $N = 3114$.

Gap $g = 5$. On $I = [-2, 7]$, take

$$\begin{aligned} v &= (2, 0, 4, 4, 4, 4, 1, 3, 3, 3), \\ A_5 v &= (-2, 2, 4, 2, 10, 12, 11, 12, 0, 11, 5, 6, 6, -3). \end{aligned}$$

Then $D = 96$ and $N = 764$.

Gap $g = 7$. On $I = [-2, 9]$, take

$$\begin{aligned} v &= (2, 0, 3, 4, 4, 4, 1, 3, 2, 3, 2, 3), \\ A_7 v &= (-2, 2, 3, 1, 10, 11, 10, 12, 1, 10, 3, 10, 4, 5, 5, -3). \end{aligned}$$

Then $D = 97$ and $N = 768$.

Gap $g = 8$. On $I = [-8, 16]$, take

$$\begin{aligned} v &= (4, 4, 4, 3, -3, 3, 9, 1, 19, 22, 21, 22, 4, 16, 8, 12, 5, 6, 0, -4, \\ &\quad -1, 1, 0, 1, 0), \\ A_8 v &= (4, 0, 8, 11, 14, 8, -7, 8, 26, 3, 53, 61, 59, 63, 9, 46, 19, 35, \\ &\quad 10, 21, -4, -8, -3, 4, 1, 1, 1, 1, 0). \end{aligned}$$

Then $D = 2487$ and $N = 19672$.

All exact comparisons for these six witnesses are collected in table 1. The third column is the raw integer from (41); the last column is the reduced or partially reduced exact difference from T .

Every integer and every rational number in table 1 is positive. Thus (39) proves $\sup \text{spec}(H_g) > c_6 + 1/250$ for these six gaps.

Table 1: Exact small-gap Rayleigh comparisons.

g	N/D	$\Delta(N, D)$	$N/D - T$
1	812/97	5598897096603523	5598897096603523/1212500000000000
2	866/109	484843129173031	484843129173031/1362500000000000
3	3114/393	702232566651387	234077522217129/1637500000000000
5	764/96	587568260556064	18361508142377/375000000000000
7	768/97	98897096603523	98897096603523/1212500000000000
8	19672/2487	174815250030533	174815250030533/3108750000000000

2.3 One witness for every $g \geq 9$

On $I = [-2, 11]$, use the fixed vector

$$v_* = (4, 0, 7, 8, 8, 9, 1, 7, 3, 6, 1, 4, 1, 2), \quad \|v_*\|^2 = 391. \quad (42)$$

Only coefficients τ_i with $-4 \leq i \leq 11$ enter its nonzero image. The right defect can therefore affect the calculation only for $g = 9$ or $g = 10$; all $g \geq 11$ have the same local coefficient word. On $J = [-4, 13]$, direct evaluation gives

$$A_9 v_* = (-4, 4, 7, 3, 20, 24, 23, 24, 5, 19, 11, 15, 6, 10, 5, 5, 3, -2), \quad (43)$$

$$A_{10} v_* = (-4, 4, 7, 3, 20, 24, 23, 24, 5, 19, 11, 15, 6, 10, 5, 5, 1, -2), \quad (44)$$

$$A_g v_* = (-4, 4, 7, 3, 20, 24, 23, 24, 5, 19, 11, 15, 6, 10, 5, 5, 1, 2), \quad (g \geq 11). \quad (45)$$

Moreover,

$$\|A_9 v_*\|^2 = 3102, \quad \|A_{10} v_*\|^2 = 3094, \quad \|A_g v_*\|^2 = 3094 \quad (g \geq 11).$$

Hence every $g \geq 9$ satisfies

$$\sup \text{spec}(H_g) \geq \frac{3094}{391} = \frac{182}{23}. \quad (46)$$

The final exact comparison is

$$\Delta(182, 23) = 10563229091557 > 0, \quad \frac{182}{23} - T = \frac{10563229091557}{2875000000000000} > 0. \quad (47)$$

Finite propagation, rather than sampling, is what reduces the infinite tail of gap lengths to the three images (43)–(45).

2.4 Completion of the single-gap hierarchy

For $g = 4$, the defect lattice is $4\mathbb{Z}$, so the operator is the period-eight reference bulk and

$$\sup \text{spec}(H_4) = \eta < c_6. \quad (48)$$

For $g = 6$, the global interface theorem gives

$$\sup \text{spec}(H_6) = c_6, \quad \dim \ker(H_6 - c_6) = 2. \quad (49)$$

The six small vectors and the fixed vector v_* exhaust every positive gap outside $\{4, 6\}$. Equation (15) transfers the conclusion between the two lifts, and reflection transfers it between the two orientations. The smallest printed margin is the $g = 8$ value

$$\frac{174815250030533}{31087500000000000} > 0.$$

We have proved the following strict statement.

Theorem 2.1 (Uniform abnormal single-gap separation). *For either lift and either interface orientation,*

$$\sup \text{spec}(H_6) = c_6, \quad \dim \ker(H_6 - c_6) = 2.$$

For every positive integer $g \notin \{4, 6\}$,

$$\sup \text{spec}(H_g) > c_6 + \frac{1}{250}.$$

Thus $G6$ is the unique minimizer among abnormal positive single gaps. The statement does not compare $G6$ with arbitrary multi-gap or finite-core interfaces.

3 Reproducibility manifest

3.1 Checkpoint and evidence boundary

This supplement is assembled against the repository checkpoint

`e365e1553ad73a8a534fb67f5ee76562521609ce`.

Every path listed below exists at that checkpoint. The JSON certificates store exact finite objects and rational enclosures. The checker chain re-evaluates their exact mathematical contracts. Two coordinate implementations separately audit the global unsquared exclusions, while some symbolic transfer and elimination routines are shared and disclosed; the package therefore does not claim a clean-room implementation of every symbolic identity. A successful checker is evidence only in combination with the analytic reductions printed in the article and in this supplement; it is not a replacement for those reductions.

No immutable external archive identifier has yet been assigned. The archive record and its persistent identifier are therefore pending and are not represented by a provisional or invented DOI.

3.2 Exact artifact inventory

Table 2: Repository artifacts used by this supplement.

Role	Repository-relative path
G6 edge certificate	<code>research/proofs/task53/certificates/g6_global_edge.json</code>
G6 local matching certificate	<code>research/proofs/task50/certificates/g6_interface_certificate.json</code>

Role	Repository-relative path
G6 elimination certificate	research/proofs/task51/certificates/c6_exact_evans_elimination.json
G6 atlas certificate	research/proofs/task53/certificates/g6_grassmann_atlas.json
G6 local matching checker	research/scripts/verify_target_a_task50_interface.py
G6 elimination checker	research/scripts/verify_target_a_task51.py
G6 edge checker	research/scripts/verify_target_a_task53_a3.py
G6 atlas checker	research/scripts/verify_target_a_task53_a2.py
Exact-2r certificate	research/proofs/task55/certificates/exact_2r_cluster.json
Exact-2r checker	research/scripts/verify_target_a_task55_exact_2r.py
Exact-2r tests	research/scripts/test_target_a_task55_exact_2r.py
Single-gap structure certificate	research/proofs/task55/certificates/single_gap_structure.json
Single-gap checker	research/scripts/verify_target_a_task55_single_gap.py
Single-gap tests	research/scripts/test_target_a_task55_single_gap.py
Single-gap integer checker	research/scripts/verify_target_a_task56_single_gap.py
Single-gap integer tests	research/scripts/test_target_a_task56_single_gap.py
Uniform-separation certificate	research/proofs/task57/certificates/uniform_single_gap_separation.json
Uniform-separation checker	research/scripts/verify_target_a_task57_uniform_single_gap.py
Uniform-separation tests	research/scripts/test_target_a_task57_uniform_single_gap.py
Orders 8–30 certificate	research/counterexamples/target_a_minimality_certificate.json
Orders 8–30 checker	research/scripts/verify_target_a_minimality_certificate.py
Orders 8–30 tests	research/scripts/test_verify_target_a_minimality_certificate.py
Order-32 certificate	research/counterexamples/target_a_n32_period8_certificate.json
Order-32 checker	research/scripts/verify_target_a_n32_certificate.py
Order-32 tests	research/scripts/test_verify_target_a_n32_certificate.py

Role	Repository-relative path
Recovery-order certificate	research/proofs/task55/certificates/small_order_exact_classification.json
Recovery-order checker	research/scripts/verify_target_a_task55_small_order_exact.py
Recovery-order tests	research/scripts/test_target_a_task55_small_order_exact.py
Order-40 certificate	research/proofs/task55/TARGET_A_ORDERS_34_46_CERTIFICATES.json
Order-40 checker	research/scripts/verify_target_a_task55_orders_34_46.py
Order-40 tests	research/scripts/test_target_a_task55_orders_34_46.py
Finite-bridge certificate	research/proofs/task54/TARGET_A_TASK54_EVENTUAL_THRESHOLD_CERTIFICATE.json
Finite-bridge checker	research/scripts/verify_target_a_task54_threshold.py

The exact- $2r$ certificate binds the rank-two one-interface input, the Floquet contraction and conditioning bounds, the Gram estimate, the codimension- $2r$ complement, the exact count, and the $2r$ -dimensional Feshbach constants. Its focused suite contains tamper tests for dimension, scope, constants, dependencies, and the coordinate form of the Schur complement.

The single-gap structure certificate records transfer, Evans, quotient symmetry, and recurrence data. The Task 56 checker independently reconstructs every integer vector and its full $A_g v$ image. The uniform-separation certificate stores the threshold (40), all seven quotient comparisons, and the minimum exact margin. Its checker reconstructs the rational comparisons and rejects non-strict or altered rows. The mathematical reason that seven rows suffice is the finite-propagation argument in section 2, not an empirical scan over gap lengths.

3.3 Verification commands

From the repository root, the focused independent checks are:

```
python3 research/scripts/verify_target_a_task53_a2.py
python3 research/scripts/verify_target_a_task53_a3.py
python3 research/scripts/verify_target_a_task50_interface.py
python3 research/scripts/verify_target_a_task51.py
python3 research/scripts/verify_target_a_task55_exact_2r.py
python3 -m pytest -q \
    research/scripts/test_target_a_task55_exact_2r.py
python3 research/scripts/verify_target_a_task55_single_gap.py
python3 -m pytest -q \
    research/scripts/test_target_a_task55_single_gap.py
python3 research/scripts/verify_target_a_task56_single_gap.py
python3 -m pytest -q \
    research/scripts/test_target_a_task56_single_gap.py
python3 research/scripts/verify_target_a_task57_uniform_single_gap.py
python3 -m pytest -q \
```

```

    research/scripts/test_target_a_task57_uniform_single_gap.py
python3 research/scripts/verify_target_a_minimality_certificate.py
python3 -m pytest -q \
    research/scripts/test_verify_target_a_minimality_certificate.py
python3 research/scripts/verify_target_a_n32_certificate.py
python3 -m pytest -q \
    research/scripts/test_verify_target_a_n32_certificate.py
python3 research/scripts/verify_target_a_task55_small_order_exact.py
python3 -m pytest -q \
    research/scripts/test_target_a_task55_small_order_exact.py
python3 research/scripts/verify_target_a_task55_orders_34_46.py
python3 -m pytest -q \
    research/scripts/test_target_a_task55_orders_34_46.py
python3 research/scripts/verify_target_a_task54_threshold.py

```

The commands should be run from a clean checkout of the checkpoint printed above. No floating-point finite-ring eigensolve is required for the exact count in theorem 1.2, and no floating-point comparison is required for theorem 2.1.

3.4 Scope of the supplement

The separated-interface theorem is restricted to $r \in \{1, 2, 3\}$ and $D \geq 1040$. It counts eigenvalues with multiplicity and does not prove individual simplicity or a sign for the leading interaction. The single-gap theorem concerns positive single gaps only. Neither theorem classifies arbitrary multi-interface minimizers.